

Information Processing 1 — Week 3 exercises

Complete each exercise on your own laptop before the end of the exercise class. You can score up to 10 points by answering all the questions. It is very important to ask an instructor to check your work before the end of the class. Answering each question contributes 2 points towards your final score for the course. Answers submitted one week late will have a 50% penalty and contribute 1 point towards your final score. If you are stuck or have any questions then raise your hand and an instructor will try to help you.

1 Using Python like a calculator [2 points]

Python has the following mathematical operators:

code	result	notes
$a + b$	$a + b$	
$a - b$	$a - b$	
$a * b$	$a \times b$	
a / b	$a \div b$	result is always a <code>float</code>
$a // b$	$\lfloor a \div b \rfloor$	result is rounded down towards $-\infty$
$a \% b$	$a \bmod b$	the remainder from $a // b$
$a ** b$	a^b	if <code>a</code> and <code>b</code> are both <code>ints</code> then the result is <code>int</code>

Using these operators, write Python expressions to calculate the answers to the following problems. Evaluate the expressions in the interactive IDLE window, or write a short script that calculates and prints out all four of the answers.

1. How many seconds are there in 42 minutes 42 seconds?
2. Given $c = 2\pi r$ what is the circumference of a circle of radius 5?
3. Given $a = \pi r^2$ what is the area of a circle with radius 5?
4. Given $v = \frac{4}{3}\pi r^3$ what is the volume of a sphere with radius 5?

Hint: after executing this statement

```
from math import pi
```

the variable '`pi`' will contain the value of π .

▷▷ Show each of your four expressions to an instructor who will record your work.

2 Algorithms and programs [2 points]

Part 1. The following algorithm converts a temperature from Celsius to Fahrenheit, where: $f = \frac{9}{5}c + 32$

1. start
2. input integer temperature c
3. let $f = c * 1.8$
4. let $f = f + 32$
5. output f
6. stop

Here are the lines of a Python program that implements this algorithm.

```
f = f + 32
print(f)
f = c * 1.8
c = int(input("c: "))
```

The lines are not in the correct order. Copy and paste the lines into a Python script, in the correct order. Run the script to make sure it works. (Test cases: 0C = 32F, 50C = 122F, and 100C = 212F)

Part 2. The following algorithm converts from Fahrenheit to Celsius, where: $c = \frac{5}{9}(f - 32)$

1. start
2. input integer temperature f
3. let $c = f - 32$
4. let $c = c * 5$
5. let $c = c / 9$
6. output c
7. stop

Implement the algorithm as a Python script. Test the script using the same test cases as above.

▷▷ Demonstrate both of your scripts to an instructor..

3 Flowcharts and programs [2 points]

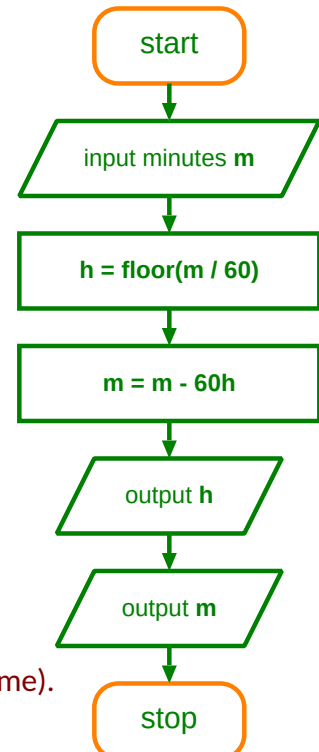
Write a program that inputs the total length of a movie in minutes and then outputs the length of the movie in hours and minutes. The flowchart is shown on the right.

To make this work you will need to find the *floor* of a floating-point number (the closest integer that is no larger than the floating-point number). There are several ways to do this, for example, by converting the `float` into an `int` or by using the `'/'` operator which performs integer division even if one or both of its operands are `floats`.

Test your program on several movie lengths in minutes that have obviously correct answers when expressed as hours plus minutes, like this (where the text that you type as input is shown in red):

```
minutes: 215
3 hours 35 minutes
```

▷▷ Ask an instructor to check and record your work now (before you run out of time).



4 Multiple inputs and outputs [2 points]

Write a program that:

1. inputs the length of the two shorter sides of a right triangle, then
2. outputs the length of the hypotenuse and the area of the triangle.

Test it with several obvious cases, for example:

```
a: 3
b: 4
hypotenuse 5.0 area 6.0
```

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

5 Vector dot product [2 points]

Python understands several other types, some of which can be used to represent vectors. Try typing these expressions into the interactive IDLE window:

```
(3, 4)
(3, 4)[0]
(3, 4)[1]
```

The `input()` function returns the text of what you type 'verbatim', as a string, without trying to interpret it as any kind of value. To interpret the text as a program value you can pass it to the `eval(s)` function, where `s` is a string and the result is the value you would obtain if you typed the contents of the string as part of a program. Try this in the interactive IDLE window or in a script:

```
s = input("vector: ")
v = eval(s)
print(v[0])
print(v[1])
```

and type the input `"(3, 4)"` (without the quotes).

You now know enough to write a program that prompts for two vectors, A and B , and then prints their dot ('inner') product $A \cdot B$, where: $A \cdot B = A_x B_x + A_y B_y$

Your program should behave like this:

```
vector A: (3,1)
vector B: (3,-1)
(3, 1) . (3, -1) = 8
```

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

6 Challenges [1 bonus point]

In case you found the first five questions too easy, here are some more challenging exercises. If you finish all of the challenges before the end of the class then you will earn 1 bonus point towards your final score.

6.1 Simple 'while' loops

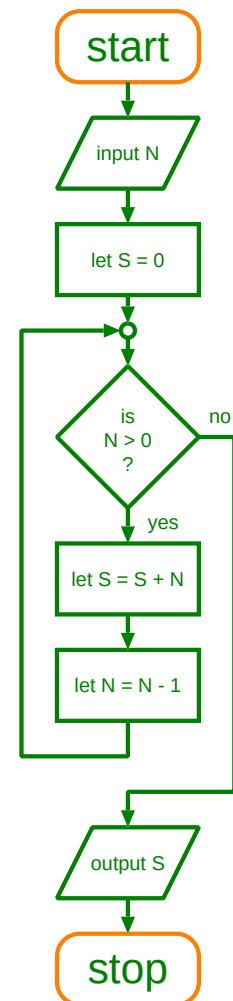
This Python program prints the first ten integers, from 10 down to 1.

```
N = 10
while (N > 0):
    print(N)
    N = N - 1
print("done")
```

Note how indentation is used to indicate which statements are inside the 'body' of the loop, just like we have been doing for pseudo-code.

The flowchart on the right shows a variation on this program that calculates the sum of the first N counting numbers. Modify the program so that it implements the flowchart. Verify that the sum of the first ten counting numbers is 55:

N: 10
55



6.2 Factorial function

Make a copy of your previous program. Modify it trivially to calculate the factorial of N , $N!$, where:

$$N! = \begin{cases} N \times (N-1)! & \text{if } N > 1, \\ 1 & \text{otherwise} \end{cases}$$

(Assume the input will never be < 1 .) Verify the program using an easy input value, e.g:

N: 5
120

6.3 Fibonacci function

Make a copy of your previous program. Modify the program so that it prints the N^{th} Fibonacci number, F_N , where: $F_0 = 0$, $F_1 = 1$, and $F_N = F_{N-1} + F_{N-2}$. The sequence of Fibonacci numbers is therefore:

0 1 1 2 3 5 8 13 21 34 55 89 144 ...

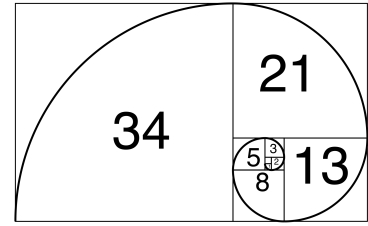
Test your program by printing the 13th Fibonacci number, F_{13} , which should be 233.

6.4 Limit of the ratio of successive Fibonacci numbers

As the Fibonacci numbers become larger the ratio between them $\frac{F_n}{F_{n-1}}$ rapidly tends towards a constant. Use your program to find the value of that constant to a reasonable accuracy.

6.5 A famous ratio

Use your favourite Internet search engine to find the name of the famous mathematical constant that you discovered in the previous question. (Typing the first four digits, with decimal point, into a search engine should find the answer for you instantly.)



▷▷ Show your challenge solutions to an instructor and claim your bonus point.